

二元函数的极值

设函数 $f(x, y)$ 在区域 G 内有定义, P_0 是 G 的内点, 存在 P_0 的某去心邻域 $D = \overset{\circ}{N}_\varepsilon(P_0) \subset G$, 使得当 $P \in D$ 时, 恒有 $f(P) \leq f(P_0)$ (或 $f(P) \geq f(P_0)$)

则称 $f(P_0)$ 为函数 f 在 G 上的极大值 (或极小值)

极值 极值点 严格极大值

极值的必要条件

设 $f(x, y)$ 在 (x_0, y_0) 可偏导, 且 $f(x_0, y_0)$ 是函数 f 的极值, 则 $f'_x(x_0, y_0) = f'_y(x_0, y_0) = 0$

驻点

若 $f(x, y)$ 在 (x_0, y_0) 可偏导, 且 $f'_x(x_0, y_0) = f'_y(x_0, y_0) = 0$

则称 (x_0, y_0) 为函数 f 的驻点

可疑极值点

将驻点与不可偏导的点称为可疑极值点

极值判别法 I

设 $P_0(x_0, y_0)$, $G = N\Delta(P_0)$, f 在 G 内连续, 且在 $U = \overset{\circ}{N}\Delta(P_0)$ 内连续可微

$$\text{记 } \mu(x, y) = f'_x(x, y)(x - x_0) + f'_y(x, y)(y - y_0)$$

若 $\forall (x, y) \in U$, $\mu(x, y) > 0$, 则 $f(x_0, y_0)$ 为极小值

若 $\forall (x, y) \in U$, $\mu(x, y) < 0$, 则 $f(x_0, y_0)$ 为极大值

二元函数拉格朗日公式

$$f(x, y) = f(x_0, y_0) + f'_x(\xi, \eta)(x - x_0) + f'_y(\xi, \eta)(y - y_0)$$

$$\xi = x_0 + \theta(x - x_0) \quad \eta = y_0 + \theta(y - y_0) \quad (0 < \theta < 1)$$

$$\text{有 } f(x, y) - f(x_0, y_0) = (f'_x(\xi, \eta)(\xi - x_0) + f'_y(\xi, \eta)(\eta - y_0)) \cdot \frac{1}{\theta}$$

$$= \frac{1}{\theta} \mu(\xi, \eta) \quad 0 < \theta < 1$$

对 $\forall (x, y)$, $\mu(x, y) > 0$ 则 $f(x, y) - f(x_0, y_0) > 0$. 故 $f(x_0, y_0)$ 为极小值

极值判断法 II 设 $P_0(x_0, y_0) \in \mathbb{R}^2$, $G = N\Delta(P_0)$, 函数 f 在 G 内二阶连续可微, 且 $f'_x(x_0, y_0) = f'_y(x_0, y_0) = 0$,

$$\text{令 } A = f''_{xx}(x_0, y_0), \quad B = f''_{xy}(x_0, y_0), \quad C = f''_{yy}(x_0, y_0)$$

若 $B^2 - AC < 0, A > 0$ 则 $f(x_0, y_0)$ 为极小值

若 $B^2 - AC < 0, A < 0$ 则 $f(x_0, y_0)$ 为极大值

若 $B^2 - AC > 0$, 则 $f(x_0, y_0)$ 不是极值

$f(x, y)$ 在 (x_0, y_0) 处的一阶泰勒展开

$$f(x, y) - f(x_0, y_0) = f'_x(x_0, y_0)h + f'_y(x_0, y_0)k + \frac{1}{2} (f''_{xx}(\xi, \eta)h^2 + 2f''_{xy}(\xi, \eta)hk + f''_{yy}(\xi, \eta)k^2),$$

$$h = x - x_0, k = y - y_0. \quad \xi = x_0 + \theta_1 h \quad \eta = y_0 + \theta_1 k \quad (0 < \theta_1 < 1)$$

当 $h \rightarrow 0, k \rightarrow 0$ 时

$$f''_{xx}(\xi, \eta) = A + \alpha, \quad f''_{xy}(\xi, \eta) = B + \beta, \quad f''_{yy}(\xi, \eta) = C + \gamma$$

$$\alpha, \beta, \gamma \rightarrow 0. \quad \text{令 } h = \rho \cos \theta, k = \rho \sin \theta, \quad \rho = \sqrt{h^2 + k^2}$$

$$\begin{aligned} \Rightarrow f(x, y) - f(x_0, y_0) &= \frac{1}{2} (Ah^2 + 2Bhk + Ck^2) + \frac{1}{2} (\alpha h^2 + 2\beta hk + \gamma k^2) \\ &= \frac{1}{2} (A \cos^2 \theta + 2B \cos \theta \sin \theta + C \sin^2 \theta) \rho^2 + o(\rho^2) \end{aligned}$$

$$\text{当 } \varphi(\theta) = A \cos^2 \theta + 2B \cos \theta \sin \theta + C \sin^2 \theta \neq 0$$

...

最小二乘法

设两个变量 x, y 之间的关系近似于线性函数关系. 测得 n 对数据 (x_i, y_i) , 试求

$y = a + bx$ 使得平方和 $u(a, b) = \sum_{i=1}^n (a + bx_i - y_i)^2$ 取到最小值.

称 $y = a + bx$ 为回归直线

$$\begin{cases} u'_a = \sum_{i=1}^n 2(a + bx_i - y_i) = 0 \\ u'_b = \sum_{i=1}^n 2(a + bx_i - y_i) x_i = 0 \end{cases}$$

$$\Rightarrow \begin{cases} na + (\sum_{i=1}^n x_i) b = \sum_{i=1}^n y_i \\ (\sum_{i=1}^n x_i) a + (\sum_{i=1}^n x_i^2) b = \sum_{i=1}^n x_i y_i \end{cases}$$

$$\begin{vmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{vmatrix} = n \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2 = \sum_{i=1}^n \sum_{i < j \leq n} (x_i - x_j)^2 \neq 0$$

故函数 $u(a, b)$ 有唯一驻点, 方程组有唯一解 (a_0, b_0)

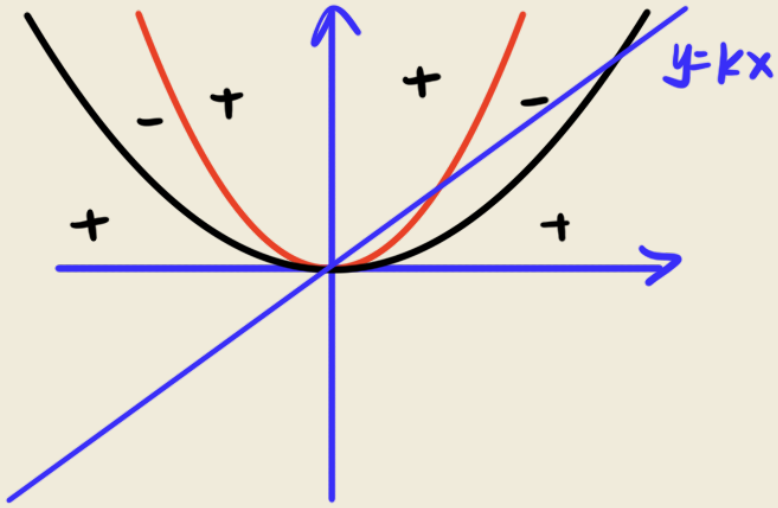
$$\begin{aligned} \text{证: } & \begin{matrix} (x_1 - x_2)^2 + (x_1 - x_3)^2 + (x_1 - x_4)^2 + \dots + (x_1 - x_n)^2 \\ (x_2 - x_3)^2 + \dots + (x_2 - x_n)^2 \\ \dots \dots \dots \\ (x_{n-1} - x_n)^2 \end{matrix} = (n-1) \sum_{i=1}^n x_i^2 - 2 \sum_{i < j \leq n} x_i x_j = \sum_{i=1}^n \sum_{i < j \leq n} (x_i - x_j)^2 \end{aligned}$$

所有射向原点的直线, 在原点处取到最小, 求在原点的邻域内, 原点的值是否取最小

理想条件, 取每条射线满足其在原点最小的邻域的最小值

但事实上有无数条直线, 无法找到最小的邻域

例如函数 $F(x,y) = (y-x^2)(y-2x^2)$



$$f(x, kx) = (kx - x^2)(kx - 2x^2)$$

当 $k \neq 0$ 时, $kx^2 + o(x^2)$ 极值取 0

当 $k = 0$ 时, $2x^4$ 极值取 0

条件极值

在曲面 S 上取 P 点, 若在 P 点取到极值, 则存在 λ

$$\nabla f(P) = \lambda \nabla \varphi(P)$$

定理: 设函数 $f(x, y, z)$ 连续可微, 函数 $\varphi(x, y, z)$ 连续可微, 函数 $f(x, y, z)$ 满足约束方程 $\varphi(x, y, z) = 0$ 的条件极值在点 $P_0(x_0, y_0, z_0)$ 取得, 且 $\varphi'_z(P_0) \neq 0$ 则

$$F(x, y, z, \lambda) = f(x, y, z) + \lambda \varphi(x, y, z)$$

$$\text{则 } P_0(x_0, y_0, z_0) \text{ 满足 } \begin{cases} F'_x = f'_x(x, y, z) + \lambda \varphi'_x(x, y, z) = 0 \\ F'_y = f'_y(x, y, z) + \lambda \varphi'_y(x, y, z) = 0 \\ F'_z = f'_z(x, y, z) + \lambda \varphi'_z(x, y, z) = 0 \\ F'_\lambda = \varphi(x, y, z) = 0 \end{cases}$$

$$\varphi(P_0) = \varphi(x, y, z) = 0$$

f 在 $P_0(x_0, y_0, z_0)$ 取得了条件的最值

$\varphi'(P_0) \neq 0$ 存在 (x_0, y_0) 的邻域 U , 使方程 $\varphi(x, y, z) = 0$ 有唯一解 $z = z(x, y)$ 并且

$$z_0 = z(x_0, y_0) \quad \varphi(x, y, z(x, y)) = 0 \quad \forall (x, y) \in U$$

$$\frac{\partial z}{\partial x}(x_0, y_0) = - \frac{\varphi'_x(P_0)}{\varphi'_z(P_0)} \quad \frac{\partial z}{\partial y}(x_0, y_0) = - \frac{\varphi'_y(P_0)}{\varphi'_z(P_0)}$$

$$\Rightarrow \begin{cases} \frac{\partial f}{\partial x}(P_0) + \frac{\partial f}{\partial z}(P_0) \frac{\partial z}{\partial x}(x_0, y_0) = 0 \\ \frac{\partial f}{\partial y}(P_0) + \frac{\partial f}{\partial z}(P_0) \frac{\partial z}{\partial y}(x_0, y_0) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} f'_x(P_0) - \frac{1}{\varphi'_z(P_0)} f'_z(P_0) \varphi'_x(P_0) = 0 \\ f'_y(P_0) - \frac{1}{\varphi'_z(P_0)} f'_z(P_0) \varphi'_y(P_0) = 0 \end{cases}$$

$$\text{记 } \lambda_0 = - \frac{1}{\varphi'_z(P_0)} f'_z(P_0) \quad \begin{cases} f'_x(P_0) + \lambda_0 \varphi'_x(P_0) = 0 \\ f'_y(P_0) + \lambda_0 \varphi'_y(P_0) = 0 \\ f'_z(P_0) + \lambda_0 \varphi'_z(P_0) = 0 \\ \varphi(P_0) = 0 \end{cases}$$

例 求 $u = x^a y^b z^c$ 满足 $x+y+z=1$ 的极值点, $(x, y, z, a, b, c > 0)$

那求 $\ln u = a \ln x + b \ln y + c \ln z + \lambda(x+y+z-1)$

$$\text{而 } \nabla L = 0 \quad \begin{cases} F'_x = \frac{a}{x} + \lambda = 0 \\ F'_y = \frac{b}{y} + \lambda = 0 \\ F'_z = \frac{c}{z} + \lambda = 0 \\ F'_\lambda = x+y+z-m=0 \end{cases}$$

$$\Rightarrow (x, y, z) = \frac{1}{a+b+c} (a, b, c)$$